

Supply Chain Network in Business Analytics

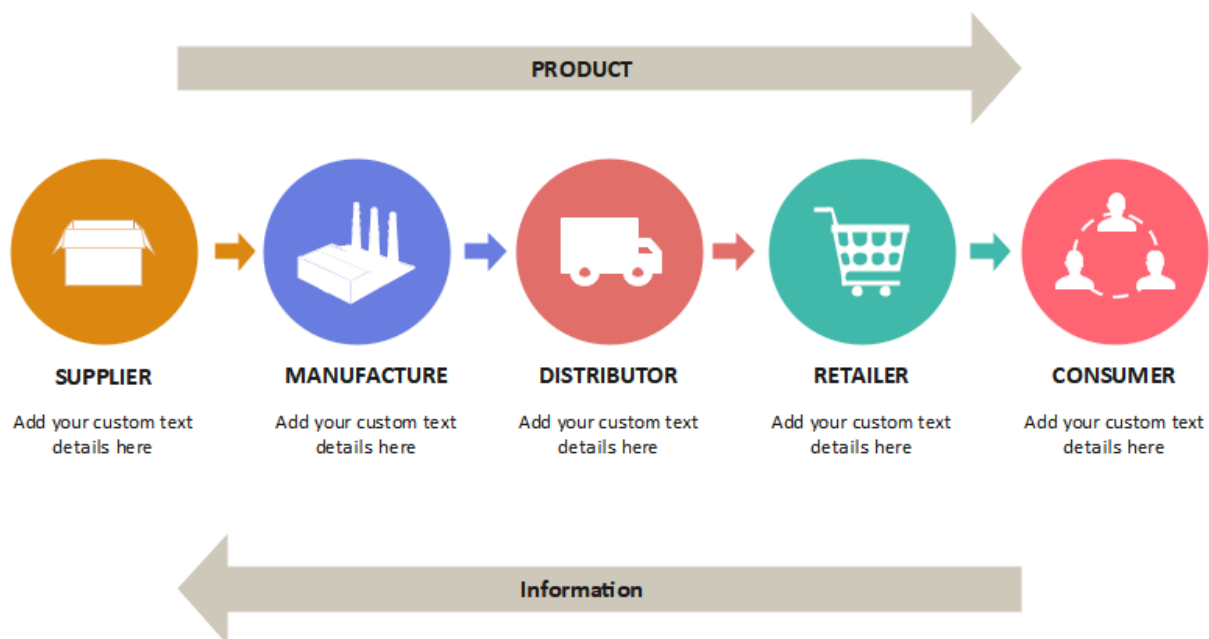
A **Supply Chain Network** is a system that connects suppliers, manufacturers, warehouses, transporters, retailers, and customers for the smooth flow of materials, products, information, and money from the starting point to the final customer.

In **Business Analytics**, supply chain networks are analyzed using data, technology, and decision-making methods to improve efficiency, reduce cost, and increase customer satisfaction.

Example:

- A company like Amazon uses analytics to track orders, delivery time, warehouse stock, and customer demand.

Components of Supply Chain Network



1. Suppliers

Suppliers provide raw materials or parts needed for production.

Example: Cotton supplier for a textile company.

2. Manufacturers

Manufacturers convert raw materials into finished products.

Example: Mobile phone companies assembling smartphones.

3. Warehouses

Warehouses store products before distribution.

Functions:

- Stock maintenance
 - Product safety
 - Easy transportation
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4. Distribution Centers

These centers help move products quickly to retailers or customers.

5. Retailers

Retailers sell products to customers. **Example:** Supermarkets, Online stores

6. Customers

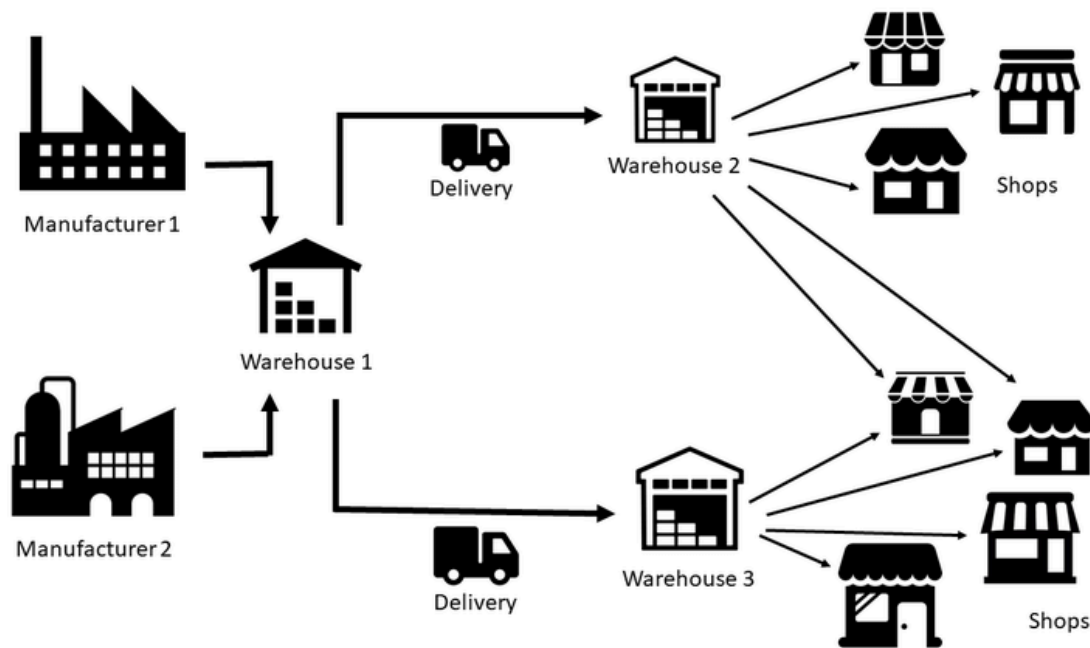
Customers are the final users of the products. Customer demand is very important in supply chain planning.

Role of Business Analytics in Supply Chain Network

Business analytics helps companies make data-driven decisions.

- 1. Demand Forecasting** - Analytics predicts future customer demand using past sales data.
 - 2. Inventory Management** - Helps maintain the correct amount of stock.
 - 3. Transportation Optimization** - Analytics finds the best delivery routes and transport methods.
 - 4. Supplier Performance Analysis** - Companies analyze supplier quality, delivery time, and reliability.
 - 5. Risk Management** - Analytics identifies risks such as:
 - Delivery delays
 - Natural disasters
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Operational Workflow of Supply Chain Network



Types of Supply Chain Networks

1. Centralized Supply Chain - All operations are controlled from one main center.

Advantages

- Easy management
- Better coordination

2. Decentralized Supply Chain - Operations are managed from multiple locations.

Advantages

- Faster response
- Better local service

Benefits and Applications of SCN

Benefits	Applications
1. Reduces overall cost 2. Increases customer satisfaction 3. Improves delivery speed 4. Better decision making	● E-commerce companies ● Manufacturing industries ● Healthcare industries ● Food delivery services ● Automobile industries

Case Study: Applying HR Analytics to Predict Demand for Hourly Employees for One Year

HR Analytics is the process of collecting, analyzing, and interpreting employee-related data to improve human resource decisions.

In this case study, HR analytics is used to predict the future demand for hourly employees for one year.

Many organizations require hourly employees for daily operations such as:

- Customer service
- Manufacturing
- Retail sales
- Delivery services

If company hires:

- too many employees, labor cost increases.
- fewer employees, work efficiency decreases.

Therefore, HR analytics helps organizations predict the required number of hourly employees accurately.

1. Company Background

Suppose a retail company wants to forecast the demand for hourly employees for the next year.

The company faces:

- Seasonal customer demand
- Employee turnover
- Weekend sales increase

The management wants to:

- Reduce labor shortages
 - Improve workforce planning
 - Control employee costs
-

2. Business Problem

The company has the following problems:

1. Employee shortage during peak seasons
 2. Excess employees during low sales periods
 3. Increased overtime costs
 4. Customer service delays
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3. Data Collection

The HR department collects historical data such as:

- Previous employee records
- Monthly sales data
- Festival and holiday schedules

Example: Walmart uses workforce analytics for employee scheduling and staffing.

4. Data Preparation - The collected data is cleaned and organized.

Activities include:

- Removing duplicate records
 - Correcting missing values
 - Organizing monthly employee data
 - Standardizing formats
-

Factors Affecting Employee Demand

1. Higher sales require more employees.
 2. Festival seasons increase customer traffic.
 3. Employees leaving the company increase hiring needs.
 4. Longer working hours require more workers.
 5. More customers require additional support staff.
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Analytical Method Used

The company uses **Predictive Analytics** and **Forecasting Techniques**.

Methods used:

- Regression analysis
- Time-series forecasting
- Trend analysis

Forecasting Process

Step 1: Analyze Historical Data

Past employee demand patterns are studied.

Step 2: Identify Trends

The company identifies:

- Peak sales months
- Low demand periods

Step 3: Build Prediction Model

A forecasting model is created using historical data.

Step 4: Generate Predictions

The model predicts monthly employee demand for the next year.

Example Forecast Table

Month	Predicted Employees Required
January	120
February	115
March	130
April	140
Festival Season	180

Benefits of HR Analytics in Employee Forecasting

1. Better workforce planning
 2. Reduced labor shortages
 3. Lower overtime costs
 4. Improved employee scheduling
 5. Better customer service
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Solutions

1. Use real-time HR data
 2. Update forecasting models regularly
 3. Improve employee retention
 4. Use automated scheduling systems
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Applications of HR Analytics

- Employee performance analysis
- Workforce planning
- Recruitment analytics
- Employee retention analysis
- Salary planning

Example:

- Amazon uses HR analytics for workforce management and staffing optimization.
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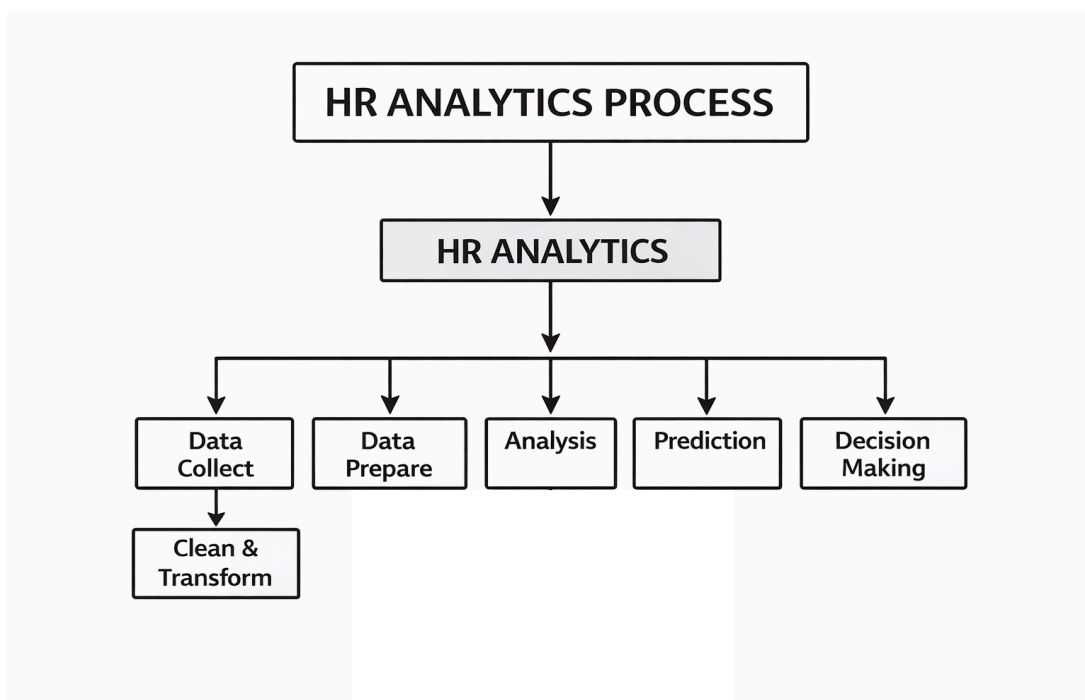
Analytics in HR for Recruitment, Training, Performance Evaluation and Attrition Prediction

Analytics in HR

Analytics in HR means using data, statistical methods, and models to make better decisions about employees. It helps organizations to understand employee behavior, improve performance, and predict future outcomes.

According to the notes, analytics follows a process:

- ❖ Data collection
- ❖ Data preparation
- ❖ Analysis
- ❖ Prediction
- ❖ Decision making



1. Analytics in Recruitment

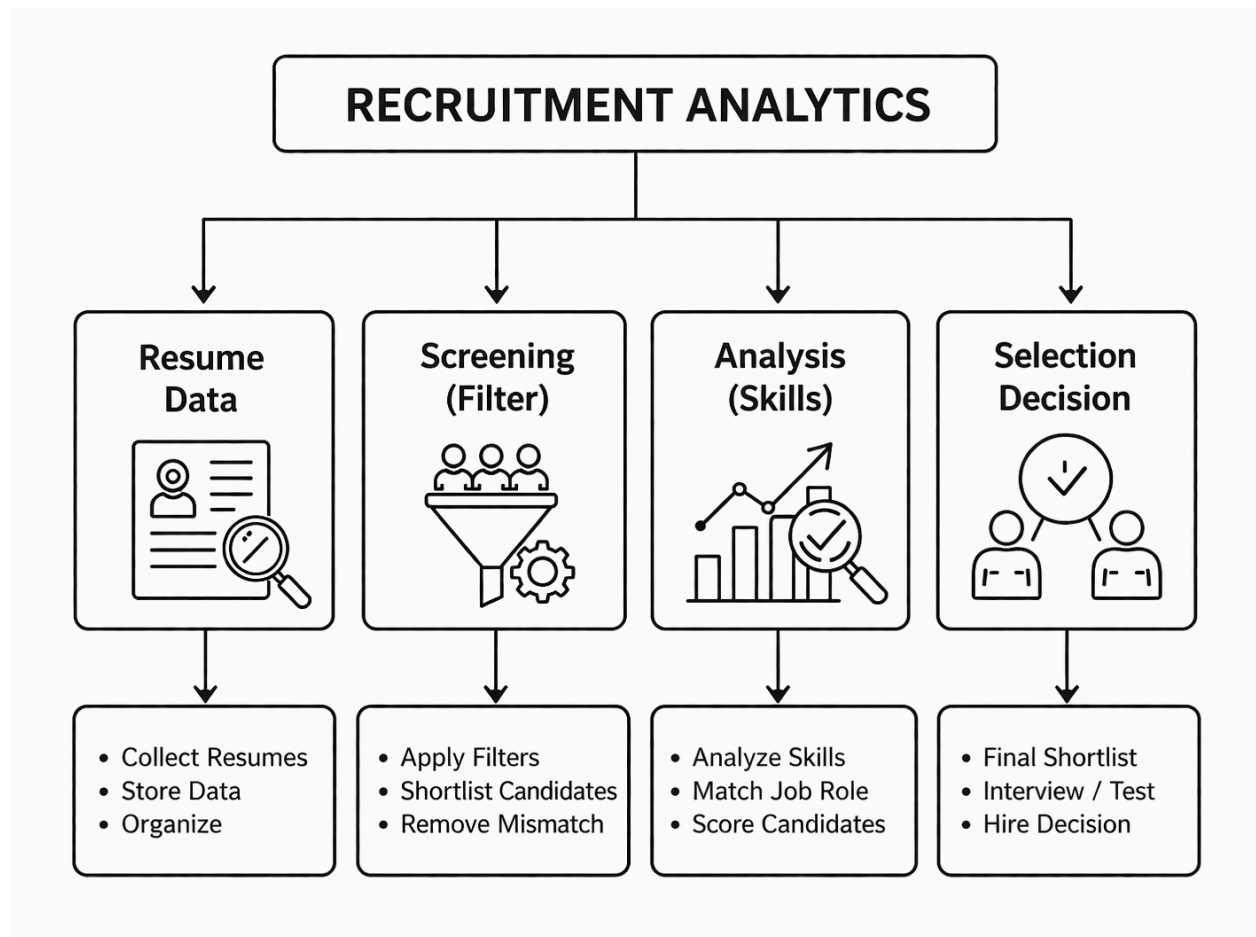


Fig 5.1 Using data to identify and select the best candidates.

Example:

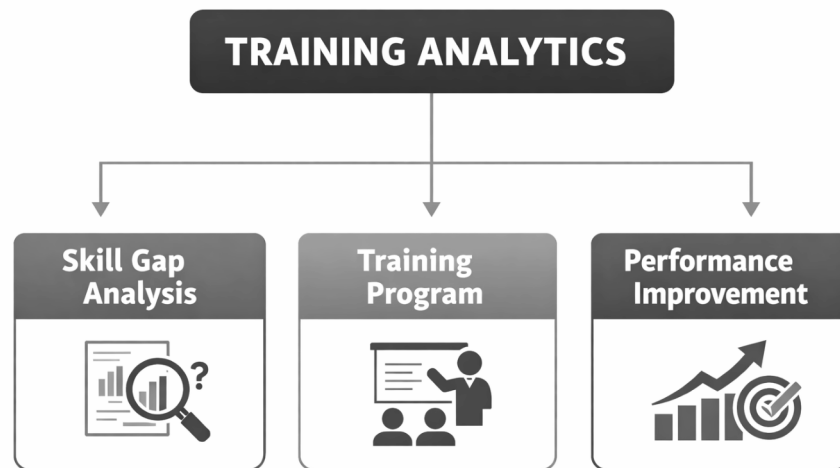
- ❖ If data shows candidates with certain skills perform better, company prefers those skills

Benefits:

- ❖ Faster hiring
- ❖ Better candidate selection

2. Analytics in Training and Development

Utilizing data-driven insights to refine employee competencies and enhance the efficacy of instructional programs.



- ❖ Use diagnostic analytics → determine root causes of low productivity

Example:

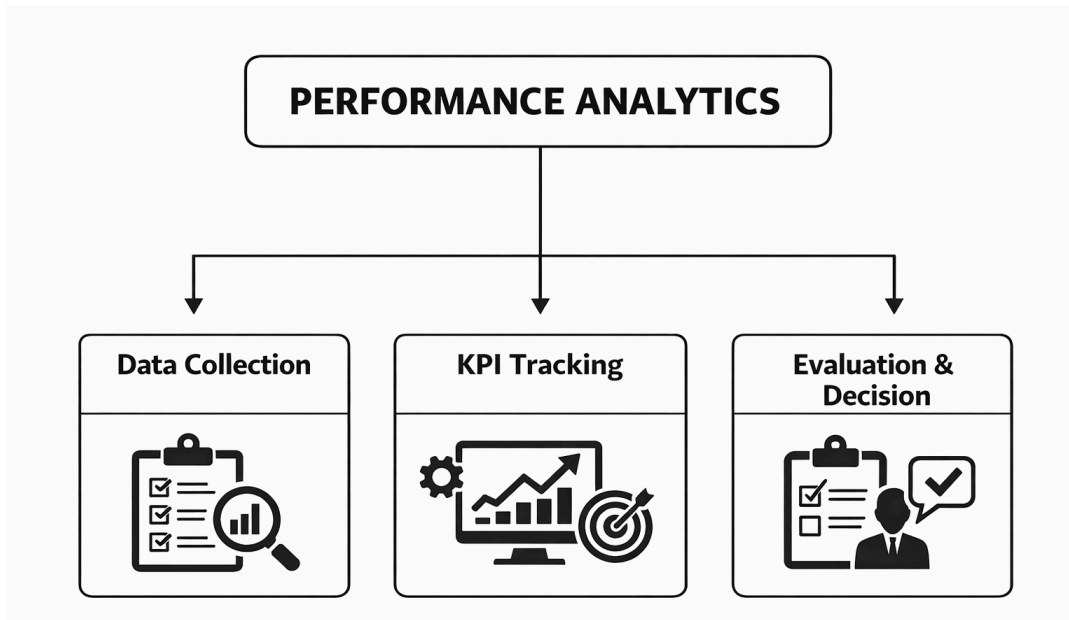
- If assessments identify a weakness in communication, specific training is allocated to that domain

Benefits:

- ❖ Enhances workforce skillsets
- ❖ Optimizes instructional expenditures
- ❖ Boosts overall organizational productivity

3. Analytics in Performance Evaluation

Implementing data-driven methodologies to quantify and monitor employee performance levels.



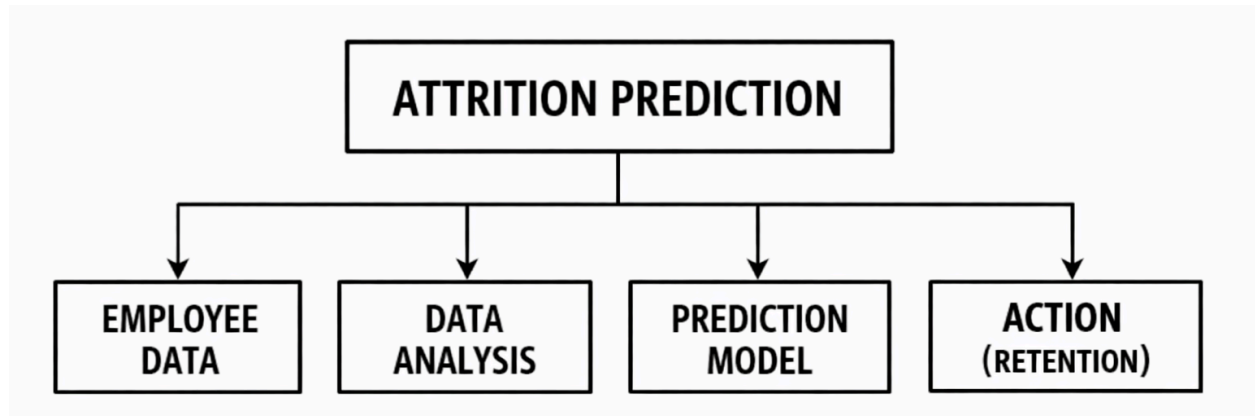
- ❖ Apply diagnostic analytics → identify causes of performance

Benefits:

- ❖ Ensures objective and fair evaluations
- ❖ Supports data-backed promotion and compensation decisions
- ❖ Increases employee engagement and motivation

4. Analytics in Attrition Prediction

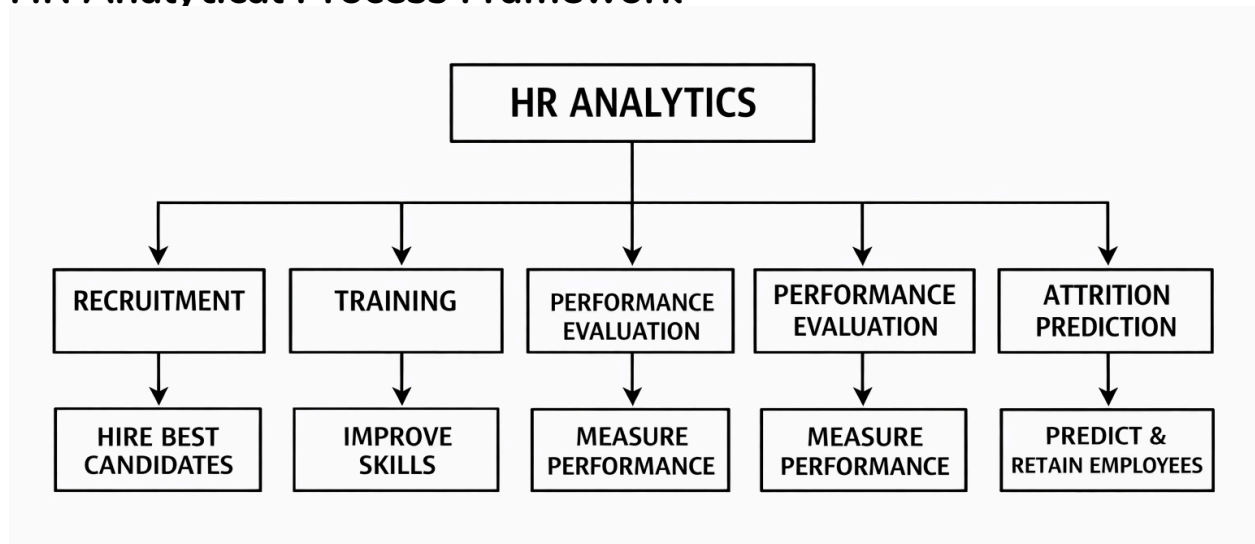
Forecasting personnel at risk of departure to optimize retention strategies.



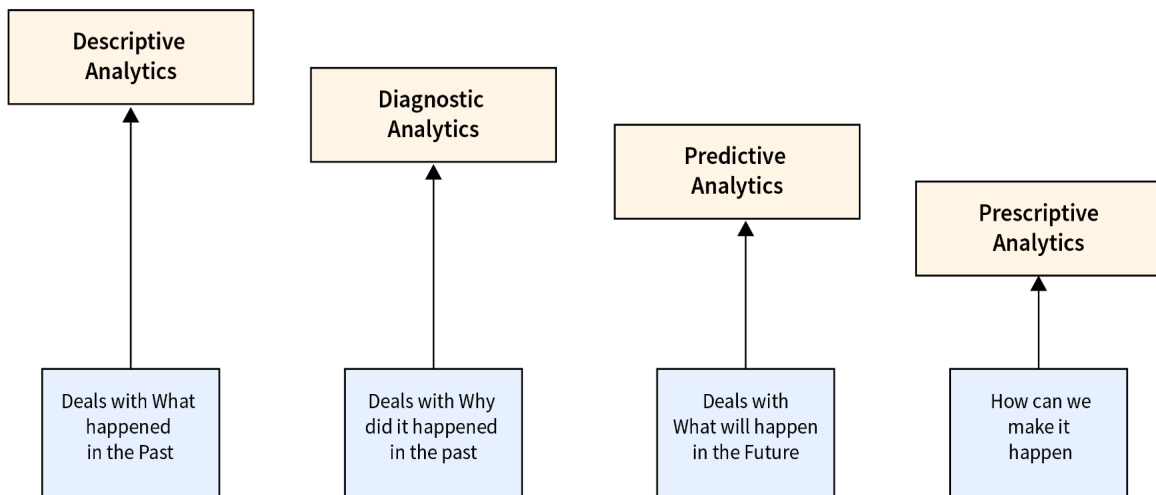
Benefits:

- ❖ Reduces employee turnover rates
 - ❖ Minimizes recruitment and onboarding costs
 - ❖ Strengthens overall employee retention
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HR Analytical Process Framework



Types of Analytics Used in HR

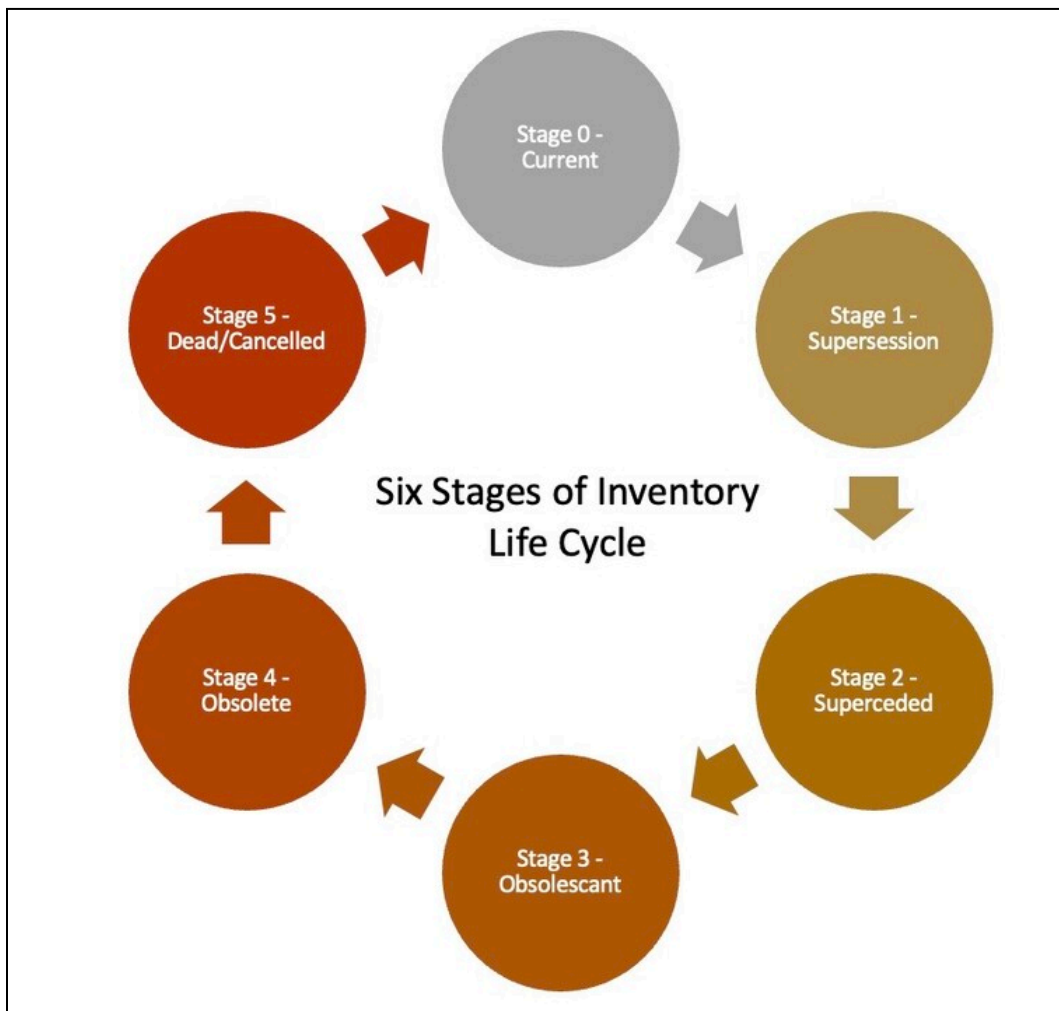


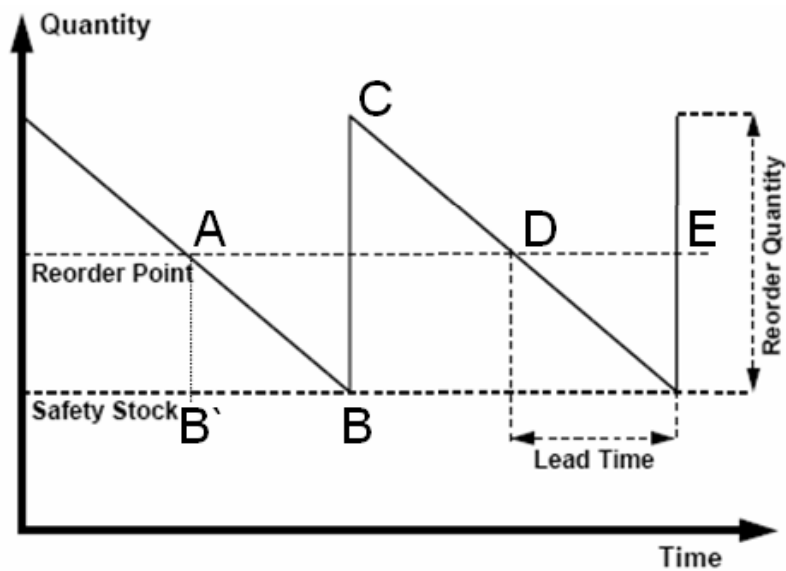
Inventory Management and Logistics

1. Analytics in Inventory Management

Inventory analytics assists in determining **optimal stock levels** and **precise reorder schedules**.

Inventory Operational Flow





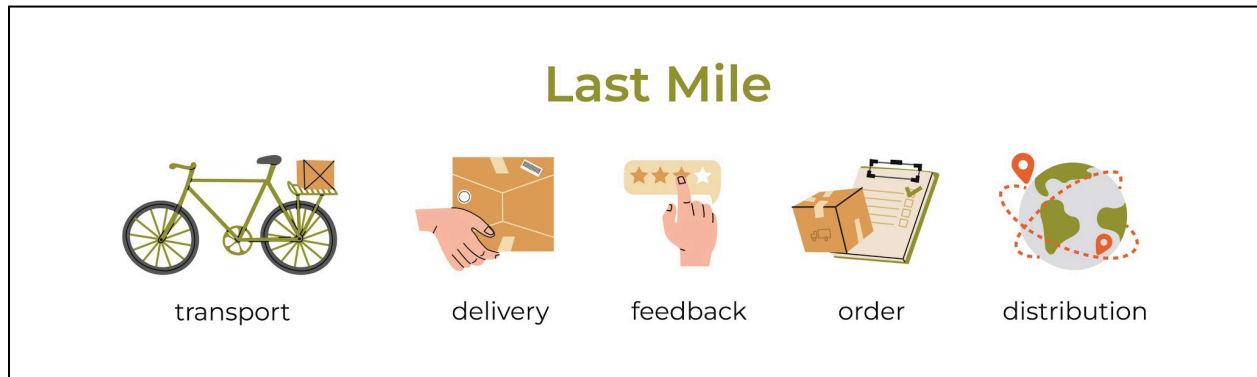
Methodologies

- ❖ Economic Order Quantity (EOQ) modeling
- ❖ ABC Classification Analysis
- ❖ Safety stock quantification
- ❖ Reorder point analytical evaluation

2. Analytics in Logistics

Logistics entails the strategic **coordination of transport and commodity distribution.**

Logistical Workflow



Analytical Contributions

- Execute shortest-path route optimization
- Mitigate fuel consumption and transportation expenditures
- Enhance delivery velocity and temporal accuracy
- Implement real-time shipment monitoring and tracking

Methodologies

- Vehicle Routing Problem (VRP) analytical modeling
- GPS integration with real-time analytical monitoring
- Simulation modeling architectures
- Advanced optimization algorithmic frameworks